

Spot The Difference: AI, Please Make This for Me!

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The Spot The Difference!

Find difference in images

Method 1

Image0 | Image1 | Image2 | Image3

Method 2

Image4 | Image5 | Image6 | Image7

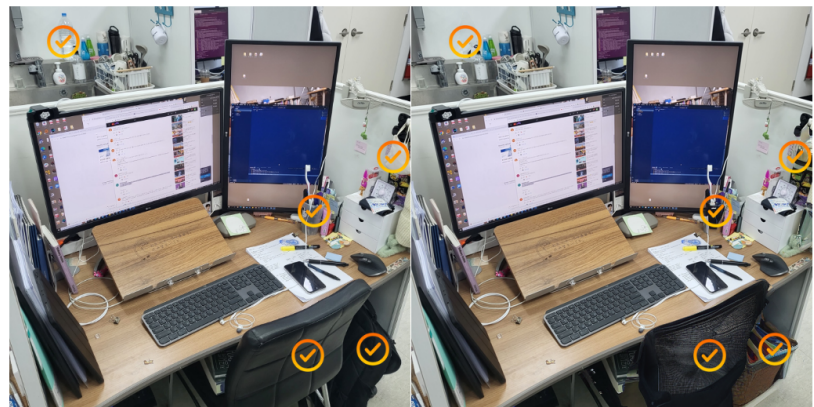


Figure 1: "Spot-The-Difference" game powered by Human and AI

ABSTRACT

Play and games are an inseparable part of our lives, and we have been playing various games from our childhood to adulthood. One of the widely recognized games is "Spot-the-Difference", which has been played and employed in a variety of domains, such as education, training, and entertainment. However, designing a custom "Spot-the-Difference" game for a certain domain requires consideration in terms of the level of game difficulty, game interest, and human intervention, therefore, is non-trivial. In this paper, we propose a novel framework based on Human-AI collaboration to automatically generate images for the "Spot-the-Difference" game. We employ Maskformer and Inpainting Stable Diffusion to automatically identify and re-draw a set of regions in the image. Finally, we conducted a user study with 19 participants to evaluate our framework. From the experimental result, we found that AI-generated

game images were enjoyable, but there is still room for improvement in the quality of the overall game playing.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence; • Applied computing → Computer games.

KEYWORDS

Generative AI, Game, Instance Segmentation, Image Generation

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1 INTRODUCTION

It is impossible to separate play and games from life. In TED talks [6] by Peter Gray, a research professor of psychology at Boston College, he said "from a biological evolutionary perspective, play is nature's means of ensuring that young mammals, including young human beings, acquire the skills that they need to acquire to develop successfully into adulthood". In addition, as humans grow, play and games are employed as a tool for entertainment and education

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naturally and utilized in improving relationships with friends and relatives [1, 5, 14]. In other words, through various games and plays, people can acquire the development of children, improve the quality of life, and achieve enhancement of brain function [1, 5, 14].

"Spot-the-Difference" is one of the most famous traditional games that people have enjoyed since childhood. The "Spot-the-difference" game does not require any computer setup and network connectivity, can be played with just a book, and has a low entry barrier due to its simple rules. Fig.1 is a snapshot of the "Spot-the-Difference" game implementation used in our study (<https://hidden-catch.web.app/>). As shown in Fig.1, the "Spot-the-Difference" presents a pair of images that have several different parts. The goal of this game is to carefully examine both images to find different parts using the player's own eye-scanning and matching strategies. It is generally told that experienced players who have abilities to rapidly understand the images can quickly recognize and spot the differences. In addition, research suggests that beginners can also improve their view of understanding images by enjoying this game continuously. Therefore, learning by "Spot-the-Difference" game has been applied to various domains, such as child education [9], clinical skill improvement for figuring out the anomaly of disease [15], and organic chemistry for figuring out the different molecular structures to understand Molecular geometry [2]. In this process, a teacher needs to configure the "Spot-the-Difference" game from their lecture materials or another reliable source in order to effectively give domain knowledge to students. This is non-trivial, however, worth the time and effort. By enjoying the "Spot-the-Difference" game made by the teacher, learners can broaden their view in terms of understanding domain-specific knowledge. However, the level of game difficulty, game interest, and human intervention should be carefully considered when collecting and creating images for the "Spot-the-Difference" game for a specific domain. Here, the level of game difficulty represents how easily the differences can be recognized by the player. The level of game interest refers to how much interest players would show when playing the "Spot-the-Difference" game, while the level of human intervention indicates how easily a user can design the game using a selected image. To maximize the benefits of a domain-specific "Spot-the-Difference" game, the aforementioned properties must be met, which is very challenging.

In this paper, we propose a novel framework for humans and AI to collaboratively design the "Spot-the-Difference" game to address the aforementioned issues. The general workflow of the proposed method is as follows: 1) a user prepares the source image, 2) the user provides the prepared image and the number of image areas to manipulate as AI input, 3) AI generates a pair of "Spot-the-Difference" game images, 4) the user verifies if the game image was generated well, 5) repeat step 2-4 until the quality of an image pair is good enough, 6) finally the user plays the "Spot-the-Difference" game made with AI-generated images. Through these steps, human and AI can co-design their own "Spot-the-Difference" game using user-selected images. More specifically, we leverage 1) instance segmentation models [3, 4, 10, 16] to detect the regions of the image to be modified and 2) generative models [7, 8, 12, 13] to create the alternative pixels for the selected areas. The details of these models are explained in Section 2. Finally, to evaluate the completeness of the proposed framework for the design of "Spot-the-Difference"

based on Human-AI Co-design, we set up the experimental process and conducted a user study.

2 MAKE THE SPOT-THE-DIFFERENCE GAME WITH AI!

In this section, we explain how the image for the "Spot-the-Difference" game (Fig. 3) is automatically generated by Human-AI collaboration. As shown in Fig. 2a, the process of designing the "Spot-the-Difference" game image is divided into two steps: 1) select a set of image regions to convert (blue box in Fig.2a). 2) generate "Spot-the-Difference" game image using Inpainting Stable Diffusion model [12] (black box in Fig.2a). The Inpainting Stable Diffusion model is a combination of the inpainting approach where an empty area of an image is reconstructed and the Stable Diffusion model which is a popular diffusion-based text-to-image generation model.

In the first step of the proposed method, a set of image regions to be converted to make differences should be set. For this, we propose two methods: 1) Object-to-Object (O2O) method in which the system detects an object in the image and replaces it with another object, and 2) Random-to-Random (R2R) method where the system randomly selects a set of areas and randomly transform them. As for the O2O method, a set of object masks in the image are generated using Maskformer [4] pretrained on tiny-COCO dataset [11]. Maskformer is a transformer-based instance segmentation model which results in a set of segmentation masks through per-pixel classification. Specifically, given an image, Maskformer creates as many masks as it can find objects in the image. However, for the 'Spot-the-Difference' game, we only need k appropriately sized masks that can control the difficulty level of a game. Therefore, we collect the masks whose area is smaller than 1/10 and larger than 1/100 of the entire image area and then sort them by a confidence score. Among them, only top- k image masks are selected as input for the generative model. In this study, we set the default value of k as 5. Fig. 3b shows five image masks of cars, windows, and bicycles made by the O2O method. In the case of the R2R method, the mask is randomly generated as a rectangular shape whose area is less than 51 x 51 pixels and more than 10x10 pixels. Fig. 3c shows five random masks.

In the generation step, we use a set of masks generated by the previous step as input for the Inpainting Stable Diffusion model. This diffusion model generates an image object using the given text prompt and replaces the mask with a newly created object. Fig. 2b shows an example of the Inpainting Stable Diffusion process. Given input data, the model erases the mask area from the image. Afterward, the model uses an inpainting approach to fill in the erased part with a newly generated object according to the input prompt.

Fig. 3 illustrates the result of mask generation and inpainting process. With the O2O method, the system performs instance segmentation, while the image masks are randomly created with the R2R method. To improve the quality of the output image, only a single pair of a mask and a prompt are taken as input at once. Therefore, to generate k different areas from the source image, it is required to repeat the process k times.

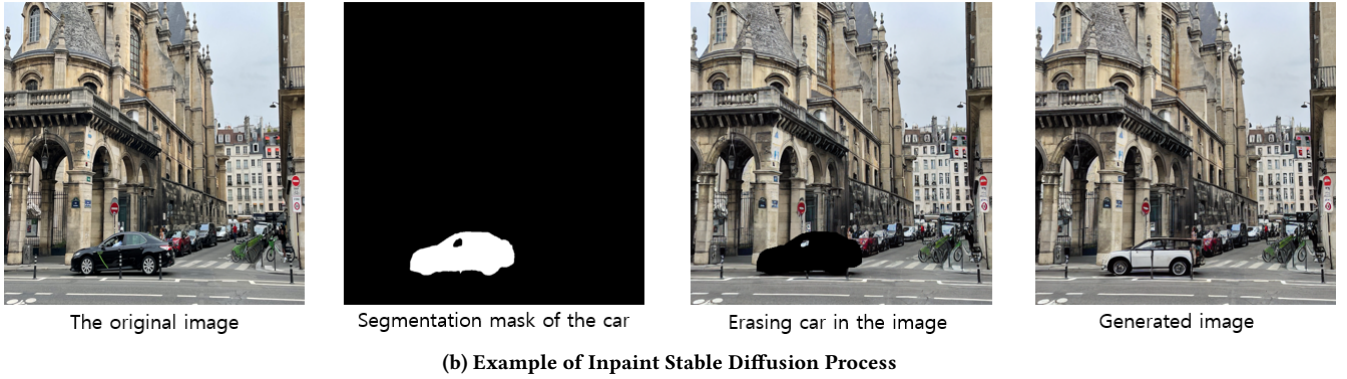
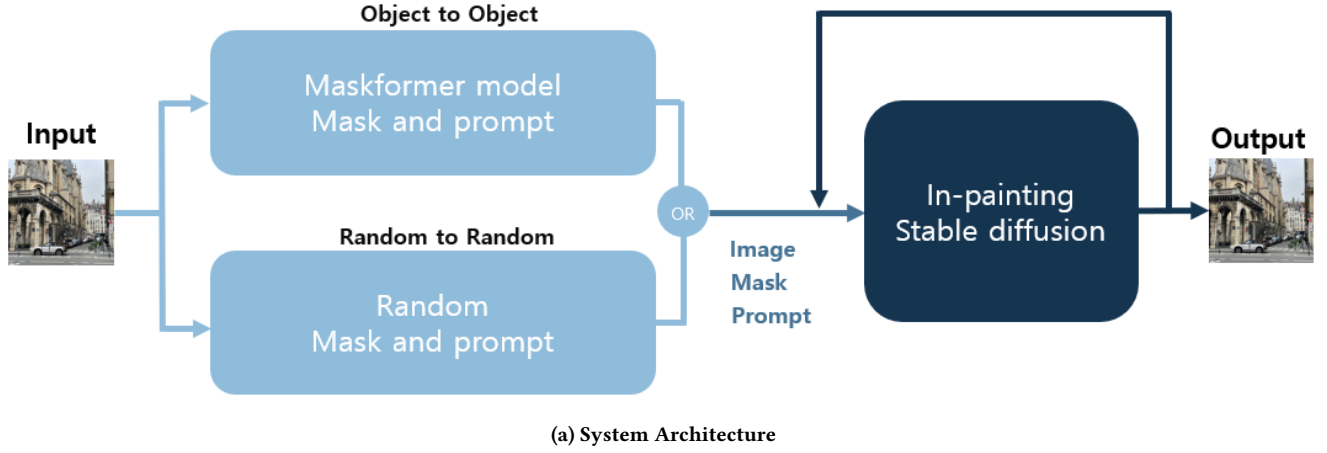


Figure 2: Workflow of the Proposed Approach

3 EXPERIMENT

3.1 Experimental process

The proposed framework of the generation process in Section 2 allows us to generate "Spot-the-Difference" images. The evaluation of the proposed approach was made through the following steps: First, we employ three images selected by the experimenter in advance and one image received from the participant as the game images. These images are used as input for the generative model described in Section 2. The model then creates a total of 8 pairs of the original and AI-generated images, consisting of four pairs from the O2O method and another four pairs from the R2R method. Second, the participants are instructed to play a web-based "Spot-The-Difference" game (<https://hidden-catch.web.app/>) implemented for our study. The web page consists of eight buttons as shown in Fig. 1. If a participant clicks a button, its corresponding image pair is displayed on the right side of the page. Lastly, after participants complete all of the stages provided in the game, they are asked to answer the research questions. Additionally, we did not inform the participants of how we generated the images (i.e., O2O or R2R approach) so as not to introduce bias in participants' answers.

3.2 Research question

The 5-point Likert scale research questions were designed to figure out user experience with the proposed approach in terms of the level of game difficulty, level of game interests, and the intention to use our framework in the future. The followings summarize our research questions:

- RQ1: "Are you satisfied with the difficulty level of the presented "Spot-the-Difference" game?" : We expect that the right level of difficulty in "Spot-the-difference" will keep people focused and interested in the image. During the experiment, we asked the participants to evaluate the difficulty of a game they played to figure out how different image generation methods affect the level of difficulty.
- RQ2: "Are you satisfied with the quality of the generated image?": The quality of the images generated by the proposed framework is one of the most important factors because it would largely affect the overall quality of the presented game. Therefore, we asked the participants to mention how they felt about the game with AI-generated images.
- RQ3: "If there is a program that enables the automatic creation of "Spot-the-Difference" game images, are you willing

to use it?": If the user feels uncomfortable while playing the game, they will not use the proposed framework anymore. Thus, we evaluated the usability of the Human-AI implementation of the "Spot-the-Difference" game from various perspectives.

3.3 Participant

To answer the aforementioned research questions, we recruited 16 university students (11 male and 5 female) and conducted a user study. The average age of the participants was 25 (SD: 2.47). Additionally, to figure out how the "Spot-the-Difference" game powered by Human-AI collaboration affects parent-child play, we also recruited three parent-child groups: father-daughter, mother-son, and mother-daughter. Their age was (40F-7D), (36M-5S), and (42M-10D), respectively. The experiment took a total of 10-15 minutes per person including gameplay and survey.

4 USER STUDY

4.1 Are you satisfied with the difficulty level of the presented "Spot-the-Difference" game?

A participant's satisfaction with the difficulty level of this game was evaluated to answer RQ1. As discussed above, the Likert-5 questionnaire was used for assessing how satisfied participants were with the difficulty of the game. As a result, we could receive an average score of 3.84 for the game based on the images generated by O2O and 3.32 for the game based on the images generated by R2R. This means that the game with images from R2R was more difficult than the game with images from O2O. In particular, 2 and 5 of 19 participants gave scores less than 2 to the game based on O2O and R2R, respectively, mentioning that the game was very difficult and challenging. It is also interesting that all the children participants gave less than two points for the game with the images generated by R2R, which implies that the game based on R2R was particularly difficult for children. Several participants commented like *"It was really hard for children to play 'Spot-the-Difference' with images generated by R2R. My daughter almost gave up finding different parts!", "I realized that some image pairs have only unnoticeable differences which make it almost impossible to complete the game; for example, only the color of a very small part was modified a little bit!"*.

We could observe that when the diffusion model used in our approach generates a new drawing, the prompt might not match the object that contains the detected mask, resulting in a just random object. Moreover, even much worse problem occurs when using the R2R method. That is, the model tends to merely perform inpainting procedure rather than generating a new object for some cases. As a result, the generated image and the original image become very similar, so participants can not easily spot the difference. We believe this could be a major reason why the participants gave much lower points for the R2R-based game. Thus, analyzing the participants' comments, we found that satisfaction with difficulty was not dependent on how the mask is generated.

4.2 Are you satisfied with the quality of the generated image?

We asked the participants to evaluate the quality of the AI-generated objects and areas in the image used in our game. We separately evaluated the quality of the output for both Basic (prepared by the research team) and Custom (received from the participants) images. As described in Fig. 4b, for Basic images, the average score is 3.63 for images generated by O2O and 3.73 for images generated by R2R. On the other hand, for Custom images, the average score is 3.89 for images generated by O2O and 3.94 for images generated by R2R as shown in Fig. 4c. The results showed no significant difference in points received for quality between images generated by O2O and R2R, and both were of good quality. However, we could observe that participants generally responded the quality of the output from Custom images was better than that from Basic images. We could also receive some negative comments regarding the quality of images used in the gameplay. In particular, one child participating in the survey even valued 1 point (very unsatisfied!) for both methods. For O2O-based game images, participants generally commented that the image pair was not clearly created: *"The boundaries of transform parts in the image were so blurry.", "The background around the target object to be transformed was modified as well so that it was very confusing/annoying when marking the difference."* In the case of the R2R approach, we manipulated the source image with a set of rectangle-shaped masks, which might make the difference between the images quite easy to be noticed. The participants also commented like *"The quality of the transformed area was much lower than the rest of an image, so it easily caught the eye."*

To sum up, we could conclude that the quality of images generated through diffusion models, used in the presented game, needs to be improved: several participants pointed out that, *"Image resolution needs to be improved.", "Some converted images look a little bit unnatural."*; one participants from child group also commented like *"The image was really weird!"*

4.3 If there is a program that enables the automatic creation of "Spot-the-Difference" game images, are you willing to use it?

This section summarizes the comments and feedback from the participants regarding RQ3. As shown in Fig. 4d, the average score regarding the intention to use the proposed approach is 3.47 for the O2O-based game and 3.73 for the R2R-based game, respectively. We also ask participants to explain the reasons for such a choice. We found that the responses regarding the intention to use the proposed method can be classified into the following categories: level of game interests and engagement in creating images.

First, those who mentioned the level of game interests loved our prototype since it allows one to utilize an image familiar to oneself, one can feel like they are designing and participating in the game. The participants from the child group also commented that they really enjoyed making the game using their own photos depicting the places they went to in the past and what they had done before. Thus, generating the "Spot-the-Difference" game with images received from participants drew positive responses. Second, in terms of engagement in creating images, participants said *"If the images for the 'Spot-the-Difference' game can be created*



Figure 3: Example of Generated Images for Game Play

automatically, it would be really comfortable! I don't want to make it one-by-one myself". Also, participants from the parent group said they are willing to use it if they could easily create games that could engage interested children.

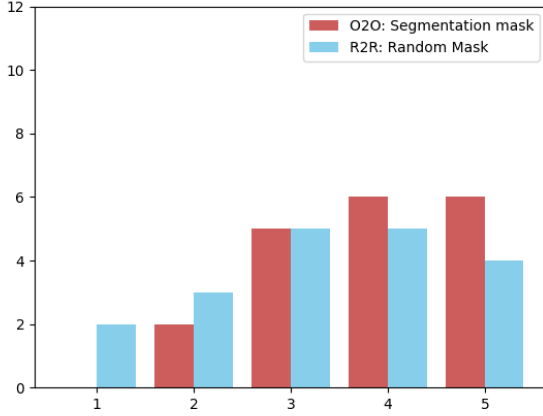
5 DISCUSSION

We received several feedback from participants that the proposed approach was useful, but the image generation method still has room for improvement. As for the result of RQ1, we could observe that the generated images can affect the difficulty of a game in some cases. Specifically, the R2R method produced more natural images than O2O but does not have much variation in the images, which can make the game more challenging. On the other hand, the use of the O2O method generally resulted in a lot of variation in the image, whose quality is, however, still low. As the result of RQ2, although the scores for the image quality of both Basic and Custom images were high, both had several negative comments on generated image quality. In this regard, we had one child participant who gave very low scores and did not find the game interesting due to the very low quality of the image generated. To summarise the results, there was a trade-off between the difficulty of a game and the quality of the generated image. If the image was generated

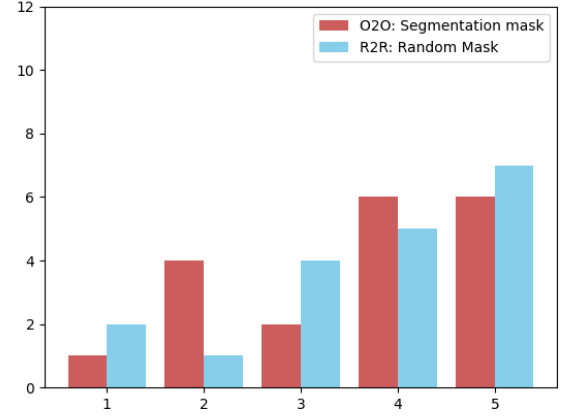
naturally then the quality increased while the difficulty tended to increase. Conversely, if the newly-generated objects in the image were distinguishable enough from the other objects, the difficulty of the game generally decreased. Therefore, our future work will focus on enhancing the proposed approach through the following improvements: i) the quality of generated images, and ii) adding more enjoyment.

5.1 Improving the quality of generated images

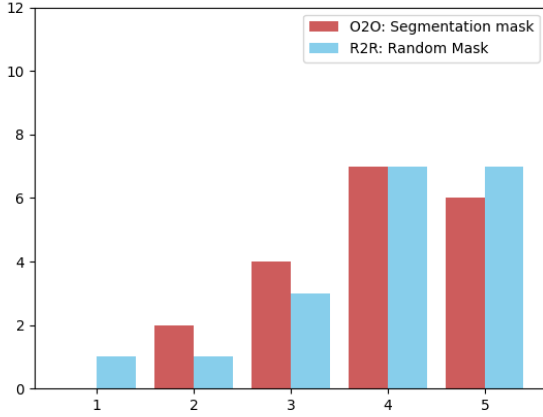
With respect to image quality, we had several issues that need to be addressed. First, the current implementation of Inpainting Stable Diffusion could not create new objects larger in size than the input mask area, which would decrease the overall quality of the game image. In order to overcome this limitation, the proposed framework should employ the models trained through objects with various sizes and resolutions. Second, faces and the human body, in general, were not generated correctly. For example, if the Diffusion model attempts to re-draw the region containing a person, a completely different person may be generated instead of the existing one. For this reason, we excluded the people masks from our experiments with the O2O-based approach, but with R2R, it was impossible to exclude regions containing people in advance



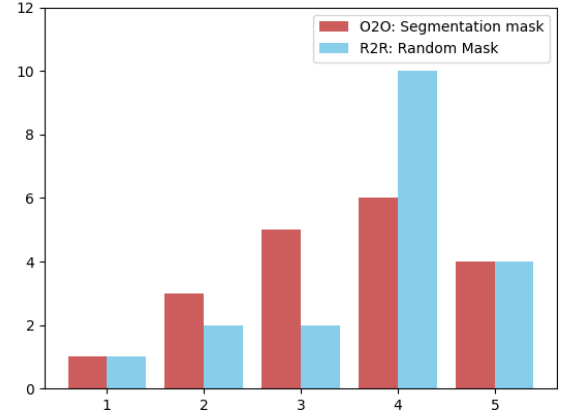
(a) Difficulty level of Spot the difference game



(b) Quality of generated images (Basic)



(c) Quality of generated images (Custom)



(d) Intention to use the proposed framework

Figure 4: Results of User Study

since the masks were selected randomly. This led to the problem of distorted people or objects being generated when the R2R method was used. Participants commented on this, stating that *"the shape of the person was sometimes shaky."* and *"there was an unnatural appearance of a person; for example, one without a neck."* To solve these problems, we also need to consider not just replacing one person with another, but also retargeting the same person to have a different pose.

5.2 Adding more enjoyment

In order to encourage users to use the proposed framework, we need to add the controllable content that users can customize. By doing so, the users may enjoy the "Spot-the-Difference" game that better suits their taste. Specifically, we define a kind of additional functions in terms of game-play, convenience, and extra contents. In terms of the game-play, we serve a controller that can manipulate

level of difficulty with just one click or drag only. In addition, we offer a hint service for beginner or even he/she is experienced player. In the event of the convenience, additional features for uploading an image from the user using web interface and masking pixels of an image where they want can be added. Finally, in the case of the extra contents, to encourage the user participation more, we can add some competitive contents such as challenges, rank systems, and time limitation.

6 CONCLUSION

In this paper, we proposed a novel framework for humans and AI to collaboratively design one of the most famous games, "Spot-the-Difference", and evaluated the proposed approach through a user study from three perspectives: level of game difficulty, level of game interest, and human intention. From the user study, we could find that there are still several limitations that should be addressed in

the future although the proposed method was suitable for making the "Spot-the-Difference" game more joyful. It is expected that the future version of the proposed framework can be applied to various domains, not only limited to "Spot-the-Difference" games, and we plan to demonstrate various use cases in our next step.

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